

Concept of Operating System

Assignment ESE

1. Define Operating System. Explain the components and role of an OS as a resource manager.
2. Differentiate between user view and system view of an Operating System with suitable examples.
3. Differentiate between Multitasking and Multiprogramming.
4. With the help of a diagram, describe the batch operating system and the multiprogramming operating system.
5. Explain various operations performed by an Operating System.
6. Explain the concept and advantages of multiprogramming operating system.
7. What is a real-time operating system? Explain hard and soft real-time systems with examples.
8. Compare and contrast multiprogramming and multitasking operating systems.
9. Explain the concept of time-sharing operating system. Write its advantages and disadvantages.
10. What are the features, advantages, and disadvantages of Android Operating System?
11. What is process management in OS? Describe process management activities.
12. What is Memory Management in OS?
13. Write a short note on:
 - (a) Command Line Interface
 - (b) Graphical User Interface
14. State any four types of system calls.
15. Define: Process, PCB
16. What is a process? Draw a process state diagram and describe each state.
17. Explain PCB with suitable diagram
18. What is a Scheduler? List and Explain any one type of it.
19. State any four criteria in CPU scheduling
20. What is memory partitioning? State their types..
21. With the help of a diagram describing a linked list
22. Compare User level thread and Kernel level Thread.
23. What is a thread? Explain the concepts of single-threaded and multithreaded programming. Also, explain the concept of user-level and kernel-level threads.
24. How to prevent Deadlock? Explain in brief.
25. Calculate average waiting time for following data using First Come First Served (FCFS) and Shortest Job First Pre-emptive (SJF) algorithms.

Process	Burst Time	Arrival Time
P0	08	0
P1	04	1
P2	05	2
P3	03	3

26. Solve a given problem by using the FCFS scheduling algorithm. Draw the correct gantt chart and calculate average waiting time and turnaround time.

Process	Burst Time	Arrival Time
P1	10	0
P2	29	1
P3	3	2
P4	7	3

27. Calculate average waiting time for following data using First Come First Served (FCFS) and Round Robin scheduling (Time Slice=3)

Process	Burst Time	Arrival Time
P0	10	0
P1	04	1
P2	07	2
P3	06	3
P4	05	4

28. Calculate average waiting time for Shortest Job First Pre-emptive (SJF) and round robin algorithms table(time slice= 4ms)

Process	Burst Time	Arrival Time
P1	10	0
P2	4	1
P3	9	2
P4	6	3

29. Explain Long-Term Scheduler, Medium-Term Scheduler ,Short-Term Scheduler.
30. Explain IPC with a diagram in detail.
31. What is deadlock? Describe necessary conditions leading to deadlock.
32. What is deadlock? Describe the necessity conditions leading to deadlock. Discuss any one method to deadlock prevention.
33. What is Thread? Explain it's types also with neat labelled diagram
34. Describe variable partitioning technique of memory management with example.
35. Explain need of memory management technique.
36. With suitable diagram, describe the concept of memory partitioning techniques.

37. Explain fixed and variable partitioning with advantages and disadvantages.

38. Describe free space management techniques:

Bit Vector

Linked List

39. Explain virtual memory and its benefits.

40. Explain Bitmap, Vector, and Linked list free space management techniques with their advantages and disadvantages

41. Explain segmentation with diagram and working of segment table.

42. Explain paging and address translation mechanism with hardware diagram.

43. Explain demand paging with valid-invalid bit mechanism.

44. Explain the steps involved in handling a page fault.

45. Explain swapping in detail with diagram.

46. Explain FIFO, LRU and Optimal page replacement algorithms with one example each.

47. Compare FIFO, OPT and LRU page replacement algorithms.

48. Given reference string:

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 2, 3, 0, 6, 7, 0, 1

Find number of page faults for:

- FIFO
- LRU
- Optimal

49. Explain difference between paging and segmentation.

50. Explain fragmentation and its types with examples.

51. What is compaction? Why is it needed?

52. Why paging suffers from internal fragmentation?

53. Why segmentation suffers from external fragmentation?

54. Explain dynamic relocation using base register.

55. Explain logical vs physical address space with diagram.

56. Why do we need virtual memory?

57. Explain difference between paging and segmentation.

58. Explain fragmentation and its types with examples.

59. What is compaction? Why is it needed?

60. Why paging suffers from internal fragmentation?

61. Why segmentation suffers from external fragmentation?

62. Explain dynamic relocation using base register.

63. Explain logical vs physical address space with diagram.

64. Why do we need virtual memory?
65. Explain the linked and indexed file allocation method with a suitable diagram
66. Explain difference between paging and segmentation.
67. Explain fragmentation and its types with examples.
68. What is compaction? Why is it needed?
69. Why paging suffers from internal fragmentation?
70. Why segmentation suffers from external fragmentation?
71. Explain dynamic relocation using base register.
72. Explain logical vs physical address space with diagram.
73. Why do we need virtual memory?
74. Explain difference between paging and segmentation.
75. Explain fragmentation and its types with examples.
76. What is compaction? Why is it needed?
77. Why paging suffers from internal fragmentation?
78. Why segmentation suffers from external fragmentation?
79. Explain dynamic relocation using base register.
80. Explain logical vs physical address space with diagram.
81. Why do we need virtual memory?
82. Explain sequential and direct access methods.